

RESEARCH ARTICLE SUMMARY

ARTIFICIAL INTELLIGENCE

AI can help humans find common ground in democratic deliberation

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INTRODUCTION: Democracy, at its best, rests upon the free and equal exchange of views among people with diverse perspectives. Collective deliberation can be effectively supported by structured events, such as citizens' assemblies, but such events are expensive, are difficult to scale, and can result in voices being heard unequally. This study investigates the potential of artificial intelligence (AI) to overcome these limitations, using AI mediation to help people find common ground on complex social and political issues.

RATIONALE: We asked whether an AI system based on large language models (LLMs) could successfully capture the underlying shared perspectives of a group of human discussants by writing a "group statement" that the discussants would collectively endorse. Inspired by Jürgen Habermas's theory of communicative action, we designed the "Habermas Machine" to iteratively generate group statements that were based on the personal opinions and critiques from individual users, with the goal of maximizing group approval ratings. Through successive rounds of human data collection, we used supervised fine-tuning and reward modeling to progressively enhance the Habermas Machine's ability to capture shared perspectives.

To evaluate the efficacy of AI-mediated deliberation, we conducted a series of experiments with over 5000 participants from the United Kingdom. These experiments investigated the impact of AI mediation on finding common ground, how the views of discussants changed across the process, the balance between minority and majority perspectives in group statements, and potential biases present in those statements. Lastly, we used the Habermas Machine for a virtual citizens' assembly, assessing its ability to support deliberation on controversial issues within a demographically representative sample of UK residents.

RESULTS: Group opinion statements generated by the Habermas Machine were consistently preferred by group members over those written by human mediators and received higher ratings from external judges for quality, clarity, informativeness, and perceived fairness. AI-mediated deliberation also reduced division within groups, with participants' reported stances converging toward a common position on the issue after deliberation; this result did not occur when discussants directly exchanged views, unmediated. Although support for the majority position increased after deliberation, the Habermas Machine demonstrably incorpo-

rated minority critiques into revised statements. We replicated these results in a virtual citizens' assembly, additionally finding that during AI-mediated deliberation, the views of groups of discussants tended to move in a similar direction on controversial issues. These shifts were not attributable to biases in the AI, suggesting that the deliberation process genuinely aided the emergence of shared perspectives on potentially polarizing social and political issues.

CONCLUSION: This research demonstrates the potential of AI to enhance collective deliberation by finding common ground among discussants with diverse views. The AI-mediated approach is time-efficient, fair, scalable, and outperforms human mediators on key dimensions. Rather than simply appealing to the majority, the Habermas Machine prominently incorporated dissenting voices into the group statements. AI-assisted deliberation is not without its risks, however; to ensure fair and inclusive debate, steps must be taken to ensure users are representative of the target population and are prepared to contribute in good faith. Under such conditions, AI may be leveraged to improve collective decision-making across various domains, from contract negotiations and conflict resolution to political discussions and citizens' assemblies. The Habermas Machine offers a promising tool for finding agreement and promoting collective action in an increasingly divided world. ■

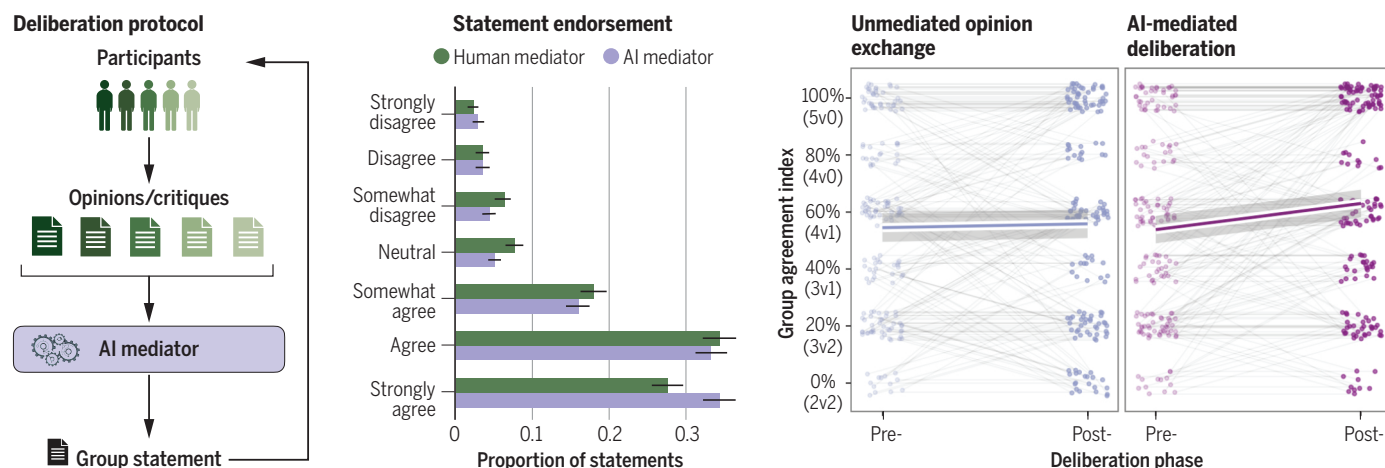
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AI helps people find common ground in collective deliberation. (Left) The AI mediator uses participants' opinions to generate group statements and iteratively refines those statements through participants' critiques. (Middle) Statements from the AI mediator (purple) garner stronger endorsement than those written by a human mediator (orange). (Right) AI mediation leaves groups less divided after deliberation, whereas simply sharing opinions with others does not.

